OMAR DHAHRI

Electromechanical Engineer

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PROFILE

Electromechanical Engineer with 4+ years of experience in industrial automation, electrical systems, and IT/network installations. Skilled in PLC/SCADA (Siemens TIA Portal, Allen Bradley Micro850), fault diagnostics, preventive maintenance, and embedded systems (STM32, C++). Experienced in SolidWorks CAD design and network infrastructure deployment, with a proven ability to improve system reliability and reduce downtime.

KEY SKILLS

- Siemens TIA Portal
- Allen Bradley Micro850 PLC
- PLC Programming
- Ladder Logic
- SCADA Systems
- Control Loops
- Industrial Automation Systems
- CAN Bus Diagnostics
- STM32 Microcontrollers
- C++ Programming
- MQTT Cloud Communication
- Embedded Systems Design
- SolidWorks CAD
- 3D Mechanical Design

WORK EXPERIENCE - SELECTED

ELECTROMECHANICAL ENGINEER (TECHNICAL SUPPORT & INSTALLATION SPECIALIST)

@ A1 Group (BULGARIA)

November 2024 - Present

- Configured and optimized enterprise-grade routers and managed switches to ensure stable LAN/WAN performance.
- Designed and deployed IP-based surveillance networks, integrating cameras with secure local storage and monitoring.
- Performed network diagnostics using packet analysis, latency testing, and signal strength evaluation.
- Implemented structured cabling and device installation to reduce connectivity failures by 15%.
- Conducted firmware updates, hardware replacements, and system performance tuning.

ELECTROMECHANICAL DESIGN ENGINEER

@T'PAP – POULINA GROUP HOLDING (TUNISIA)

April 2023 - October 2024

- Designed and optimized mechanical components, shafts, and rotating assemblies using SolidWorks.
- Developed 3D models, assemblies, and manufacturing drawings for workshop execution.

- Designed a roller conveyor system, including load analysis, material selection, and component sizing.
- Engineered and reinforced a cooling tower fan support structure with focus on vibration resistance and durability.
- Designed a vacuum pump manifold, defining piping layout and welding fabrication process.
- Supported maintenance teams through failure analysis and design improvements for reliability.

ROBOTICS & AUTOMATION INSTRUCTOR

@Engineering Geeks' Club (TUNISIA)

October 2021 - Mars 2023

- Taught students embedded systems using STM32, sensors, and motor drivers.
- Guided development of 50 autonomous robots using PID control and line-tracking algorithms.
- Led IoT-based home automation projects using microcontrollers and cloud connectivity.

INTERNSHIPS

GRADUATION PROJECT

@Les Ciments de Bizerte (Ciment Company)

Feb. 2022 – June 2022

Project: Predictive Maintenance & Failure Detection — **STM32, PLC, KPI dashboard, risk analysis**

AUTOMATION ENGINEER INTERN

@AIGK International Airport

July 201

Project: Airfield Lighting Automation System — **SCADA, control diagnostics, real-time monitoring**

CONTROL SYSTEMS INTERN

@El Gouafel Transport Company

July 2020

Project: Vehicle ECU Communication — **CAN Bus diagnostics, sensor data analysis**

ELECTRICAL TECHNICIAN INTERN

@Chiheb Electrique

August 2018

Project: Industrial Control Cabinets — **Electrical schematics, testing (multimeter & insulation)**

CERTIFICATIONS

- Allen-Bradley Micro850 PLC with IIoT – Udemy (2025)
- Siemens TIA Portal – Devices & Networks – SITRAIN (2025)
- SIMATIC – Basics of Programmable Logic Controllers (PLC) – SITRAIN (2025)
- Introduction to Data & Data Science – 365 DataScience (2024)
- Scrum Fundamentals Certified (SFC) – Scrum Study (2024)
- Scrum for Ops & DevOps (SODFC) – Scrum Study (2024)
- Negotiation Associate – Scrum Study (2024)
- Six Sigma Yellow Belt – Scrum Study (2024)
- ISO 14001:2015 Requirements – KJ Formation & Consulting (2021)
- ISO 9001:2015 Requirements – KJ Formation & Consulting (2021)
- SolidWorks CSWA – Mechanical Design – Dassault Systèmes (2017)

EDUCATION

NATIONAL SCHOOL OF ENGINEERS OF GAFSA

National Engineer Diploma in Electromechanical engineering 2019-2022

HIGHER INSTITUTE OF TECHNOLOGICAL STUDIES

Bachelor's Degree in Electrical Engineering : Automation and Industrial IT 2016-2019

LANGUAGES

- **English**
- **Arabic**
- **French**